

Solving Absolute Value Linear Equations

Algebra II Honors | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can justify why an absolute value equation splits into two cases, and solve absolute value linear equations that require multiple isolation steps.

Vocabulary

Isolate — get the absolute value expression alone before splitting.

Split into two cases — $|A| = k$ ($k > 0$) becomes $A = k$ OR $A = -k$.

Part 1 — WHY Does $|A| = k$ Split Into Two Cases?

Using Day 1's piecewise definition of absolute value, explain why $|A| = k$ ($k > 0$) means $A = k$ OR $A = -k$.

Sample: _____

Part 2 — Isolate First, Then Split

Example 1. $2|x + 1| - 3 = 7$

Isolate: $|x + 1| = \underline{\hspace{2cm}}$ Solve: $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Example 2. $5 - 2|x - 1| = -3$

Isolate: $-2|x - 1| = \underline{\hspace{2cm}}$ $\rightarrow$ $|x - 1| = \underline{\hspace{2cm}}$ Solve: $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Notice: dividing by a NEGATIVE number does not flip anything here, unlike an inequality - it just isolates the absolute value.

Part 3 — Fractions and Distributing

Example 3. $|(2x - 4)/3| = 6$

$2x - 4 = 18$ or $2x - 4 = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Example 4. $|3(x - 2)| + 1 = 10$

Isolate: $|3(x - 2)| = \underline{\hspace{2cm}}$ $\rightarrow$ $3(x - 2) = 9$ or $3(x - 2) = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

You Try. $|4(x + 1)| - 2 = 14$

$|4(x + 1)| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Practice

1. $4|x + 2| = 20$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

2. $|(x + 5)/2| = 3$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

3. $8 - 3|x - 2| = -1$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$

Exit Ticket

Solve $6 - 2|x + 3| = -4$, and explain in one sentence why the equation splits into two cases once $|x + 3|$ is isolated.

$|x + 3| = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$ _____